CedarSun Energy

Smarter Solar. Reliable Power. A Brighter Future.

Business Profile and Technical Knowledge Base

Residential solar, battery storage and monitoring in Lebanon

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CedarSun Energy is a fictional company created for EECE 503P/798S (Agentic Systems), assignment C3.
All names, contact details, figures and policies are illustrative and do not describe a real business.

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How to use this document

This document is the approved knowledge base for CedarSun Energy's virtual advisor. The advisor answers customer questions only from this document and from the short business summary. Where this document gives a **planning value** (for example, 5 peak-sun hours), the advisor may use it for a rough estimate but must say that the estimate is approximate and that a site survey is required before any final design.

Three kinds of statements appear here. **Company policy** is what CedarSun promises customers. **Illustrative specification** is a representative equipment value, not a claim about a specific manufacturer. **Engineering estimate** is a calculation that depends on assumptions and must always be presented as approximate.

Document control

Item	Detail
Document ID / version	CSE-KB-001, version 1.1
Date	October 2026
Prepared by	Layal Nassar (Head of Engineering) with Dana Mansour (Customer Experience)
Approved by	Karim Haddad (CEO)
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Intended readers	Customers (summary), the virtual advisor (full document), CedarSun sales and engineering staff

1. Company overview

CedarSun Energy S.A.L. designs, installs and supports rooftop solar and battery systems for homes and apartments in Lebanon. Our engineers size every system around the customer's actual appliances and daily routine, and our monitoring app lets customers see how their system performs after installation.

Fact	Detail
Legal / trading name	CedarSun Energy S.A.L. / CedarSun Energy
Founded	2019, Beirut
Headquarters	Sin el Fil, Beirut, Lebanon (showroom and engineering office)
Service areas	Beirut, Mount Lebanon (Metn, Keserwan, Baabda, Aley, Chouf), Jbeil, Batroun, Tripoli, Saida and Zahle. Other areas on request after review.
Customers	Private homes, villas and apartments with access to a suitable roof or terrace. We do not currently install commercial or industrial systems.
Track record	More than 380 residential systems installed since 2019
Team	About 30 staff: 8 engineers, 3 installation crews, customer support and administration
Business hours	Monday to Friday 08:00 to 17:00; Saturday 09:00 to 13:00 (Beirut time)
Contact	Email: hello@cedarsun-energy.example | Phone / WhatsApp: +961 1 000 000 (fictional)
Support line for system faults	support@cedarsun-energy.example | +961 1 000 111 (fictional)

2. Mission, values and what makes us different

Mission. To give Lebanese households reliable, understandable electricity from the sun, so that power cuts and generator dependence disrupt daily life less.

Vision. A Lebanon where every suitable home can generate and store its own clean power, designed honestly and supported for the long term.

Values

- **Honest sizing.** We recommend the system a home needs, not the largest system we can sell.
- **Transparency.** We state our assumptions, losses and limits in every proposal.
- **Safety first.** Electrical work is done only by qualified CedarSun technicians, following the relevant standards.
- **Long-term support.** A sold system is the start of the relationship, not the end.

Unique value proposition

CedarSun combines **engineer-led sizing** with **live monitoring**. Every customer receives a free site survey, a written load analysis that shows how the design was calculated, and the CedarSun Home app that shows generation, consumption and battery level in real time. Customers can see whether the system is performing as designed, and our support team is alerted automatically if it is not.

3. Services

Service	What it includes
Solar consultation and free site survey	Phone or showroom consultation, then an on-site visit (about 1.5 to 2 hours) to inspect the roof, shading, electrical panel and equipment space. No obligation.
Load analysis and system design	Review of the customer's appliances and habits, then a written design with panel count, inverter size, battery capacity and expected production.
Installation and commissioning	Mounting, wiring, inverter and battery installation, protection devices, testing, app setup and customer handover.
Battery storage	Lithium iron phosphate (LFP) batteries for evening use and backup during grid or generator outages.
Monitoring and reporting	CedarSun Home app with live data, alerts and monthly energy reports.
Maintenance and cleaning	Annual inspection plan, panel cleaning and performance checks.
Upgrades	Adding panels or battery modules to an existing CedarSun system, subject to inverter and roof capacity.

4. Equipment and technical knowledge

Values below are illustrative CedarSun specifications. Actual datasheet values appear in each customer's quotation.

4.1 Solar panels

Property	Typical CedarSun specification
Technology	Monocrystalline silicon, 450 W rated power per panel (at standard test conditions)
Efficiency	Approximately 20 to 23 percent, depending on model
Size and weight	About 2.2 m^2 and 25 kg per panel. Plan about 2.5 m^2 of usable roof per panel including spacing and walkways.
Temperature behaviour	Output falls as panels get hotter, by roughly 0.3 to 0.4 percent per degree Celsius above 25 degrees. Hot summer days reduce output compared to the rated value.
Degradation	Output declines slowly with age, about 0.4 to 0.5 percent per year
Standards	Selected from modules certified to IEC 61215 and IEC 61730

4.2 Inverters

The inverter converts the direct current (DC) from the panels into alternating current (AC) used by household appliances. CedarSun installs **hybrid inverters**, which manage the panels, the battery, the grid or generator input, and the home's loads in one unit.

Inverter rating	Typical use	Notes
3 kW	Essential loads: lighting, refrigerator, TV, internet, fans	Limited surge capacity; not for air conditioners
5 kW	Essentials plus one air conditioner	Check the surge rating against the air conditioner start-up current
8 kW	Larger homes with several high-power appliances	Appliances running at the same time must stay under the continuous rating

Configurations: **grid-tied or hybrid** systems work with the public grid or a generator as a second source, while **off-grid** systems rely on solar and batteries alone and need larger batteries. Most CedarSun residential systems are hybrid, with the grid or generator as backup.

4.3 Batteries

Property	Typical CedarSun specification
Chemistry	Lithium iron phosphate (LFP)
Module sizes	5.12 kWh nominal per module; modules combine to 10.24 kWh, 15.36 kWh and so on, up to the inverter's limit
Usable capacity	Not all nominal capacity is usable. The planning value is 90 percent depth of discharge (DOD).
Conversion efficiency	Planning value of 90 percent on the discharge path (battery to AC)
Power limit	A battery delivers a limited power (kW) as well as stored energy (kWh). A large battery cannot run an appliance that exceeds its power rating.
Standards	Selected from batteries certified to IEC 62619

4.4 Key terms customers often confuse

Term	Meaning
kW (kilowatt)	Power: how fast energy is used or produced at a moment in time
kWh (kilowatt-hour)	Energy: power over time. A 1 kW appliance running for 1 hour uses 1 kWh.
kWp	The rated peak capacity of the panel array under standard test conditions. Six 450 W panels = 2.7 kWp.
Peak-sun hours	A way to express daily sunlight: 5 peak-sun hours means as much energy as 5 hours of full-strength sun
Performance factor	Share of the theoretical output actually delivered after real-world losses (heat, dust, wiring, inverter). CedarSun uses 0.80 for planning.
Surge (start-up) power	The brief higher power some appliances need at start-up, such as compressors in refrigerators and air conditioners

5. Site assessment

Solar output depends on the site, not only on the number of panels. Every CedarSun design starts with an on-site survey, and the advisor must not give a final panel count without one. The survey checks:

- **Roof or terrace area and orientation.** In Lebanon, south-facing panels with a tilt of roughly 25 to 30 degrees generally produce the most over the year. East or west layouts are possible with lower output.
- **Shading** from nearby buildings, trees, water tanks, antennas and parapet walls, which can reduce production sharply even when only part of a panel is shaded.
- **Roof condition and structure.** Whether the roof can carry the added weight and mounting loads.
- **Electrical panel and earthing.** Condition of the existing installation, available space for protection devices, and cable routes to the inverter location.
- **Equipment location.** A ventilated, protected, accessible space for the inverter and batteries, away from direct sun and moisture.
- **Dust, salt and weather exposure.** Coastal sites need more frequent cleaning and corrosion-resistant mounting.
- **Safe access** for installation and future maintenance.

Apartments in shared buildings often need building-owner or co-owner agreement to use the roof. CedarSun can advise on what to ask but cannot grant or confirm such permission.

6. Customer load assessment

To size a system, the advisor needs to understand what the customer wants to power. It asks one or two questions at a time, in a friendly way, and does not interrogate the customer.

Intake questions

- What type of home is it (apartment, house or villa) and roughly how large is it?
- Which appliances do you use most (refrigerator, lighting, TV and internet, washing machine, air conditioner, electric water heater, oven)?
- Roughly how many hours a day does each one run?
- Do you want to power the whole home, or only essentials during outages?
- Do you use more electricity in the daytime or at night?
- Do you currently rely on a private generator, a UPS or batteries, and how many hours of public power do you get per day?
- Do you have a rooftop or terrace, and who owns or controls it?

Typical appliance values (planning use only)

These are typical figures to help a customer who does not know the wattage. The real value is on the appliance label and should be preferred when available.

Appliance	Typical running power	Typical use	Approx. energy per day	Start-up surge
LED lighting (whole home)	100 W	5 h	0.5 kWh	None
Refrigerator	150 W while running	About 8 h equivalent	1.2 kWh	2 to 3 times running power
Television and internet router	100 to 130 W	4 to 6 h	0.4 to 0.7 kWh	None
Washing machine	500 W average, up to 2,000 W while heating	1 h	About 0.5 kWh	Moderate
Air conditioner (inverter type, 1 HP)	About 800 to 1,000 W	5 h	About 4 to 5 kWh	Up to 2 to 3 times running power
Electric water heater	1,500 to 3,000 W	1 to 2 h	About 2 to 5 kWh	None, but high continuous power

Electric water heaters, ovens and several air conditioners at once are high-power loads. They usually need a larger inverter and battery, and the advisor should recommend an engineer review.

7. System-sizing methodology

The advisor can give a **rough, preliminary estimate** using the method below. It must state the assumptions and say that the final design follows the site survey. Planning assumptions: **5.0 to 5.5 peak-sun hours per day depending on area (table below)**, **0.80 performance factor**, **450 W panels**, **90 percent depth of discharge**, **90 percent discharge efficiency**.

Planning sun hours by area

Area	Planning peak-sun hours per day
Coast: Beirut, Jbeil, Batroun, Tripoli, Saida	5.0
Mount Lebanon: Metn, Keserwan, Baabda, Aley, Chouf	5.2
Bekaa: Zahle and surroundings	5.5
Area not listed or unknown	5.0 (default)

These are conservative annual-average planning values for rough estimates only. Real output depends on the exact site, season, shading and dust.

Step 1: daily energy

Daily energy (kWh) = sum of [appliance power in watts x hours of use per day] / 1000

Appliance	Power	Daily use	Energy per day
Refrigerator	150 W	8 h equivalent	1.2 kWh
Lighting	100 W	5 h	0.5 kWh
Television	100 W	4 h	0.4 kWh
Washing machine	500 W	1 h	0.5 kWh
Air conditioner	1,000 W	5 h	5.0 kWh
Total			**7.6 kWh per day**

Step 2: solar array size and panel count

Array size (kW) = daily energy / (peak-sun hours x performance factor)

Example: 7.6 / (5 x 0.80) = 1.9 kW. With 450 W panels: 1.9 / 0.45 = 4.2, rounded up to **5 panels (2.25 kWp)**.

Step 3: battery capacity and backup time

Nominal battery (kWh) = energy to store / (depth of discharge x discharge efficiency)

Example: to supply 4 kWh of essential evening energy, 4 / (0.90 x 0.90) = 4.94 kWh, so one **5.12 kWh module** is a starting point. Backup time (hours) is usable AC energy divided by average load: 4 kWh at a steady 500 W is about **8 hours**. Real runtime is shorter if the load changes or conversion losses are higher. If the customer gives no backup target, the advisor plans the battery for **40 percent of the daily energy**, which approximates evening and night use, and rounds up to whole 5.12 kWh modules.

Step 4: inverter check

The advisor asks two separate questions: how much power do appliances draw **at the same time**, and which ones have a high **start-up surge**? The inverter's continuous and surge ratings and the battery's power limit must cover both. For rough estimates the advisor assumes about **70 percent of the connected appliances run at the same time**, picks the smallest standard inverter (3, 5 or 8 kW) that covers that load, uses at least 5 kW when an air conditioner is included, and sends anything above 8 kW to an engineer as a Custom Design. A 5 kWp array does not automatically need a 5 kW inverter, and a 10 kWh battery cannot necessarily run every appliance at once.

Step 5: expected production of a given array

Daily production (kWh) = array size (kWp) x peak-sun hours x performance factor

Example: a 3.6 kWp array gives 3.6 x 5 x 0.80 = about **14.4 kWh per day** on an average day. Production is higher in summer, noticeably lower in winter and on cloudy or dusty days, and depends on shading and

orientation.

Limits of the estimate. The example above shows that 5 panels could match the home's daily energy on average. It does not prove the home can run its air conditioner continuously, cover all night-time use, or stay powered through a long outage. Only a site survey and a full load analysis can confirm those points. The advisor never promises savings, payback periods or exact output.

8. Residential packages

The packages below are **illustrative starting configurations** used to explain what is possible. They are not fixed products. Final specifications and prices come only from a written quotation after the site survey. Daily production uses the planning method in Section 7 (kWp x 5 x 0.80).

Package	Panels	Array	Inverter	Battery	Avg. daily production	Intended use
Essential Backup	4 x 450 W	1.8 kWp	3 kW hybrid	1 x 5.12 kWh	About 7.2 kWh	Lighting, refrigerator, TV, internet, fans and device charging during outages
Smart Home Hybrid	8 x 450 W	3.6 kWp	5 kW hybrid	2 x 5.12 kWh (10.24 kWh)	About 14.4 kWh	Essentials plus one air conditioner for part of the day, washing machine, evening use
High-Demand Home	14 x 450 W	6.3 kWp	8 kW hybrid	3 x 5.12 kWh (15.36 kWh)	About 25.2 kWh	Larger homes with several air conditioners and heavier daily use
Custom Design	As designed	As designed	As designed	As designed	As designed	Villas, electric water heating, cooking loads or unusual roofs; requires engineer design

Limitations of each package

- **Essential Backup:** not designed to run air conditioners or an electric water heater.
- **Smart Home Hybrid:** running an air conditioner overnight on battery alone will drain the battery quickly. The system relies on grid or generator support for heavy overnight use.
- **High-Demand Home:** needs roughly 35 m^2 of usable roof for 14 panels. Electric water heating and cooking need a separate load review.
- **All packages:** output depends on shading, orientation, season and dust. The listed production is an average planning figure, not a guarantee.

9. Installation process and timelines

Timelines are typical estimates, not guarantees. Equipment availability, weather, roof access and building approvals can change them.

Step	What happens	Typical time
1. First contact	The customer talks to the virtual advisor or the sales team and leaves contact details	Same day
2. Follow-up call	A CedarSun specialist contacts the customer	Within 1 business day
3. Site survey	Free on-site visit by an engineer or technician	Usually scheduled within 5 business days of the call
4. Design and quotation	Load analysis, system design and written quotation in US dollars	Within 3 business days after the survey

Step	What happens	Typical time
5. Approval	Customer reviews and approves the quotation; deposit paid	Customer's decision
6. Installation	Mounting, wiring, inverter and battery installation, testing	Typically 1 to 3 days for a residential system once equipment is available
7. Commissioning and handover	System tested, CedarSun Home app set up, customer shown how the system works	Same visit as installation
8. After-sales	Monitoring, support and annual maintenance	Ongoing

10. Monitoring and smart energy features

The **CedarSun Home** app is part of every installation. It shows:

- Live solar generation and total energy produced today, this month and since installation
- Household consumption
- Battery level (state of charge) and whether the battery is charging or discharging
- Whether the home is using solar, battery, grid or generator power at any moment
- Alerts for faults, low battery, or unusual drops in production (for example, heavy shading or soiling)
- A monthly energy report with simple suggestions, such as moving heavy appliances to sunny hours

Monitoring requires a working internet connection at the home. Without internet, the system still operates, but live data and alerts are unavailable until the connection returns.

11. Warranty, maintenance and support

The following are CedarSun's standard warranty terms. The quotation lists the exact terms for the selected equipment.

Item	Standard coverage
Solar panels	12-year product warranty; 25-year linear performance warranty (output at or above about 80 percent of rated power at year 25)
Hybrid inverter	10-year manufacturer warranty
Battery	10-year warranty, subject to the stated cycle and throughput limits in the datasheet
CedarSun workmanship	5 years on installation work, mounting and wiring
Not covered	Damage from misuse, unauthorized modification, floods, fire or lightning outside normal protection limits, and shading or output changes caused by new nearby construction

Maintenance

- Annual inspection and performance check, offered as a maintenance plan
- Panel cleaning, recommended every one to three months depending on dust, with more frequent cleaning near the coast or construction sites
- Customers should not climb onto the roof or open electrical equipment themselves

Support

- Fault reports by phone, email or the app: first response within 24 hours on business days
- Faults that leave the home without backup power are prioritised and answered the same business day
- Remote diagnosis is attempted first; a technician visit is scheduled if needed

12. Pricing, payment and commercial policies

- **Custom quotation.** CedarSun does not publish fixed prices. Every price is calculated after the site survey and sent as a written quotation in US dollars.
- **The advisor never states or estimates prices, discounts or payback periods.** If asked, it explains the quotation process, offers a free site survey, and logs the question for the sales team.
- **Payment terms.** Typically a deposit on approval and the balance at commissioning. Exact terms are in the quotation.
- **Financing.** CedarSun does not currently offer loans or installment plans.
- **No guaranteed savings.** Savings depend on the customer's current electricity or generator costs, which CedarSun cannot know without an assessment.
- **No invented dates or approvals.** Installation dates are agreed only after the quotation is approved.

13. Lebanon context and what we do not state

Many Lebanese households combine limited public electricity with a private generator subscription. Solar with battery storage can cover a large part of daily use and provide backup during cuts, but the right design depends on the home's loads and roof.

What the advisor can explain

- How hybrid systems work with the public grid or a generator
- What a system can and cannot run during a power cut
- How battery and solar interact when sunlight is low, such as at night or in winter
- Why a site survey is needed

What the advisor must not state, and must log instead

- Current Lebanese regulations, licensing rules, or utility approvals
- Whether a customer is entitled to export power to the grid or to net metering
- Government incentives, customs duties, or tax treatment
- Insurance or building-permit questions for a specific property
- Legal questions about roof ownership or co-owner consent
- Any detail not found in this document

CedarSun is a fictional company, and this document does not claim any official regulatory position. Questions in the list above are recorded and passed to the team to answer with verified information.

14. Team

Name	Role	Background
Karim Haddad	Co-founder and CEO	Electrical engineer with 15 years in power systems and renewable energy projects in Lebanon and the Gulf. Leads strategy and partnerships.
Layal Nassar	Co-founder and Head of Engineering	Electrical and computer engineer. Developed CedarSun's sizing method and monitoring platform; oversees all system designs.
Rami Khoury	Head of Installations	Licensed electrician with 12 years of experience. Leads the three installation crews and safety training.
Dana Mansour	Customer Experience Manager	Manages site-survey scheduling, customer follow-up and after-sales support. Designed the virtual advisor's customer journey.

15. Frequently asked questions

Q: How many solar panels do I need?

A: It depends on how much electricity your home uses, what you want to run, and your roof. The advisor can give a rough estimate from your appliances and daily hours, and an engineer confirms the final number after a free site survey.

Q: Can solar run my air conditioner?

A: Yes, in many homes, if the system is sized for it. Air conditioners use a lot of power and have a start-up surge, so the inverter, panels and battery must be chosen with them in mind. Running one overnight on battery alone needs a larger battery.

Q: What happens during a power cut?

A: A hybrid system with a battery keeps powering your chosen loads using solar and battery energy. How long it lasts depends on battery size and what you run.

Q: Will solar work on cloudy days or in winter?

A: Yes, but with lower output. Production is highest in summer and lower on cloudy days and in winter, so the grid or generator may still be needed at times.

Q: Do I still need my generator?

A: Many customers use it much less, but it depends on your loads and battery size. Heavy loads, long outages or low sun may still need a backup source. We can design the system to work alongside a generator.

Q: How long does installation take?

A: A typical residential installation takes about 1 to 3 days once the quotation is approved and equipment is available. This is an estimate, not a guarantee.

Q: How much does it cost?

A: We provide a custom written quotation, in US dollars, after the free site survey. The advisor cannot quote prices in chat.

Q: Is the site survey really free?

A: Yes. It is free and without obligation.

Q: What is the difference between kW and kWh?

A: kW is power at a moment in time. kWh is energy over time. A 1 kW appliance running for 1 hour uses 1 kWh.

Q: What batteries do you use and how long do they last?

A: Lithium iron phosphate (LFP) batteries in 5.12 kWh modules, with a 10-year warranty subject to datasheet limits.

Q: What warranties do I get?
A: Panels: 12-year product and 25-year performance. Inverter: 10 years. Battery: 10 years. CedarSun workmanship: 5 years. Exact terms are in your quotation.

Q: Can I add more panels or batteries later?
A: Often yes, depending on inverter capacity and roof space. An engineer checks this before any upgrade.

Q: Can I monitor my system from my phone?
A: Yes. The CedarSun Home app shows live generation, consumption and battery level, with alerts for faults.

Q: I live in an apartment. Can I still install solar?
A: Possibly, if you have access to a suitable roof or terrace and the building's co-owners agree. We can assess this during the site survey.

Q: Can I sell power back to the grid?
A: The advisor cannot answer questions about grid export or net metering, because this document does not contain verified information. The question is logged and the team will follow up.

Q: Do you install commercial or industrial systems?
A: Not currently. CedarSun focuses on residential systems.

16. Assistant scope, guardrails and escalation

CedarSun's virtual advisor is named **Zeina**. She is warm, clear and practical, explains technical terms in simple language, and speaks with the confidence of someone who knows solar but never pressures customers.

Message types and handling

Type of message	Examples	Handling
General question	"How does a hybrid inverter work?", "Do you offer warranties?", "How many panels would I need?"	Answer from this document. For sizing, ask the intake questions, then give a rough, labelled estimate.
Lead	"I want a quote", "Can someone visit my house?", customer shares name and email	Collect name, email and a note about the customer's needs (home type, appliances, estimate). Log with the customer-interest tool and confirm a follow-up within 1 business day.
Feedback or unanswered question	A complaint, a suggestion, or any question this document cannot answer	Log with the feedback tool. Tell the customer the question was passed to the team and offer to take contact details.

Rules the advisor follows

- Answer only from this document and the business summary. Never invent products, prices, dates, regulations or technical figures.
- Never quote prices, discounts, savings, payback periods or guaranteed output.
- Always describe sizing results as approximate and subject to a site survey.
- Never give electrical installation or repair instructions for the customer to carry out. Recommend a qualified technician.
- Never give legal, regulatory, tax or insurance advice.
- When the answer is not in this document, say so honestly, log the question, and offer a follow-up.
- Encourage interested customers to leave their name and email so a specialist can contact them, and never end a lead conversation without asking for an email.

- Stay in character as CedarSun's advisor and stay on topic: solar, batteries, energy use and CedarSun's services.

Escalation to a human

Hand the conversation to the team (by logging it and offering a callback) when the customer asks for a price or a contract, reports a fault or safety concern, asks about regulations or grid export, requests a design for a high-power load, or is unhappy with the service. For any report of smoke, burning smell, sparking or a damaged battery, the advisor tells the customer to switch the system off at the isolators if it is safe to do so, keep away from the equipment and call CedarSun's support line immediately, and to call emergency services if there is a fire.

End of document. CedarSun Energy is a fictional company created for academic use.